SENTINEL-X | ONE-PAGE PROJECT REPORT

ENTERPRISE DETECTION ENGINEERING & SOC AUTOMATION (SIEM/SOAR) PLATFORM

ACADEMIC & INDUSTRIAL REVIEW

October 2026

1. PROJECT TITLE

SENTINEL-X: Enterprise-Style Detection Engineering & SOC Automation (SIEM/SOAR) Platform

2. WHY DID YOU CHOOSE THIS PROJECT?

Modern enterprise Security Operations Centers (SOCs) are overwhelmed by millions of raw, disparate logs (Linux, Windows Active Directory, network firewalls). Manually triaging logs causes alert fatigue and inflates Mean Time to Detect (MTTD) and Respond (MTTR). I chose this project to acquire hands-on mastery in **Detection Engineering, SIEM architecture, and SOAR automation**—the primary disciplines deployed by Tier-1 defense teams (CrowdStrike, Splunk, Palo Alto Cortex). Building this platform directly bridges theoretical cybersecurity concepts with adversary behavior through the **MITRE ATT&CK;® framework**.

3. TECHNOLOGIES & TOOLS USED

Backend & Engine: Python 3.15, FastAPI (Async REST API)	**Normalization Schema:** OCSF & Elastic Common Schema (ECS)
Detection Logic: YAML Sigma rules, Sliding Time Windows	**Threat Intelligence:** Offline GeoIP, ASN, C2 DB, CMDB Asset Inventory
Database: SQLite with persistent JSON indexing	**Frontend UI:** Cyber SOC Dark-Mode Dashboard (Tailwind CSS, Chart.js)

4. HOW YOU COMPLETED THE PROJECT (APPROACH & METHODOLOGY)

- **Ingestion & Normalization:** Engineered regex & lookahead parsers converting heterogeneous logs (Linux *auth.log*, Windows EVTX *4624/4625/4688*, Zeek, Web access) into standardized *NormalizedEvent* objects.
- **Rule Authoring & Stateful Detection:** Authored 9 multi-domain rules (Authentication, Endpoint, Network) with an in-memory sliding-window manager supporting immediate triggers and thresholding (≥ 5 events / 60s).
- **Automated Context Enrichment:** Computed composite risk scores (0–100) via Threat Intelligence (Tor nodes, C2 trackers) and CMDB asset criticality (Tier-0 Domain Controller vs. Workstation).
- **SOC Incident Management:** Built automated ticketing engine assembling *INC-2026-XXXX* tickets with correlated evidence logs, visual chronological attack timelines, and MITRE ATT&CK; tags.
- **SOAR Automated Containment:** Implemented closed-loop playbooks (Edge Firewall IP drops, EDR network host isolation, Active Directory account lockout) with comprehensive audit logging.
- **Operations UI & Simulator:** Created a real-time dark-mode SOC Dashboard with live metrics, MITRE heatmap, and a 1-click Attack Simulator replaying real-world attack scenarios.

5. WHAT YOU LEARNED (KEY TECHNICAL SKILLS & CONCEPTS GAINED)

- **Detection Engineering:** Authoring and tuning production-grade YAML Sigma rules with threshold condition logic.
- **Stateful Stream Correlation:** Implementing in-memory sliding time windows and cooldowns to mitigate alert storms.
- **MITRE ATT&CK;® Mapping:** Aligning adversary techniques (T1110.001 Brute Force, T1059.001 PowerShell, T1003.001 LSASS Dumping, T1490 Ransomware Recovery Inhibiting, T1071.001 C2 Web Protocols).
- **SOAR Incident Response:** Designing closed-loop automated playbooks that slash containment response times from hours to milliseconds.

6. CHALLENGES FACED & SOLUTIONS

Challenge Encountered	Root Cause	Engineering Solution Implemented
Disparate Log Syntaxes	Heterogeneous key-value syntax across Windows EVTX, Linux syslog, and Zeek.	Engineered lookahead regex parsers extracting entities into standardized OCSF schema.
Alert Fatigue & Flooding	High-frequency attacks triggered duplicate tickets per failed attempt.	Implemented stateful sliding-window pruning with 30s entity suppression cooldowns.
Python 3.15 Compatibility	Standard PyYAML lacked pre-built binary wheels on Windows.	Integrated pure-Python StrictYAML, ensuring zero-compiler portability and stability.

7. FINAL OUTCOME

Successfully delivered a production-style, self-contained **Enterprise SOC & Detection Platform** with a 100% automated *Log → Detection → Alert → Enrichment → Incident Ticket → SOAR Containment* pipeline, a modern dark-mode Cyber Operations Dashboard running at **http://localhost:8000**, a 1-click live attack simulation center, and **11/11 automated unit and integration tests passing in under 1 second**.